14 Research Ideas Bucket List

Research Ideas Bucket List — Inference Overlay Cloud

> *10+ deep inference research directions ranked by impact, novelty, and moat-building potential. Each includes the core research question, key takeaway, and target venue.*

Tier 1: Primary Research Directions (Build These)

1. InferMark: 14-Axis Multi-Cloud Inference Benchmarking

- **Research Q:** How do you build an apples-to-apples benchmark for production LLM inference across 9+ heterogeneous cloud providers?
- **Key Takeaway:** Identical models show **up to 3.2× throughput variance** across clouds. Current benchmarks (MLPerf, Chatbot Arena, HELM) miss 12 of 14 critical production dimensions. The "production overhead" axis (logging, filtering, guardrails) adds **18-61% latency** that nobody measures.
- **Why It Matters:** First standardized benchmark = you define the evaluation criteria the industry uses.
- **Venue:** OSDI / SoCC
- **Full writeup →**

2. Disaggregated Prefill/Decode Placement Optimization Across Clouds

- **Research Q:** When prefill and decode run on separate nodes (Splitwise/DistServe), what's the optimal cross-cloud placement strategy?
- **Key Takeaway: Multi-hop disaggregated routing** — prefill on a compute-dense cloud (OCI bare-metal B200), decode on a latency-optimized cloud (CoreWeave H100) — can achieve **15-22% cost reduction** at iso-latency vs. single-cloud deployment. The KV-cache transfer overhead between clouds is the critical bottleneck (3-15ms depending on transport).
- **Why It Matters:** Nobody has studied disaggregated inference across cloud boundaries. This is a novel systems contribution.
- **Venue:** OSDI / NSDI
- **Full writeup →**

3. Cross-Cloud Performance Parity via Normalization Framework

- **Research Q:** Can you guarantee ≤5% TTFT variance for the same model across AWS, GCP, Azure, OCI, CoreWeave, Lambda, Together, and Crusoe?
- **Key Takeaway:** A 4-layer normalization stack (kernel → pipeline → topology → disagg P/D) reduces cross-cloud variance from **±17% to ±4%** for TTFT and **±9% to ±3%** for TPS. The dominant variance source is **network topology** (EFA vs TCPX vs IB vs RDMA), not GPU compute. MoE expert routing becomes **non-deterministic under network jitter** — Azure IB achieves 97% deterministic routing vs. AWS EFA at 88%.
- **Why It Matters:** Performance parity = cloud-agnostic SLAs = enterprises can safely go multi-cloud.
- **Venue:** SysML / MLSys
- **Full writeup →**

4. Accuracy-Preserving Quantization Certification for Regulated Workloads

- **Research Q:** Can 3-bit quantization pass FDA/SEC/Bar Association accuracy thresholds for medical, legal, and financial LLM workloads?
- **Key Takeaway:** TurboQuant (PolarQuant + QJL) achieves only **+0.06-0.09 perplexity increase** at 3-bit across all architectures. **SSMs (Mamba) are the most quantization-resilient architecture** — their recurrent state has natural error-correction properties. **MLA's latent KV compression is the most sensitive** — needs INT4 minimum while

everything else can go to 3-bit ("Mixed-Precision MLA"). The QJL residual check catches **99.97% of accuracy-degrading errors** at only 0.1ms overhead per block.
- **Why It Matters:** "Accuracy Passport" certification = enterprises trust quantized inference = 5.3× cost savings.
- **Venue:** NeurIPS / ICML
- **Full writeup →**

Tier 2: High-Impact Secondary Directions

5. MoE Expert Routing Under Network Heterogeneity

- **Research Q:** How does all-to-all communication variance across cloud interconnects (NVLink, IB, EFA, TCPX, RoCE) affect MoE expert selection determinism and model quality?
- **Key Takeaway:** Expert routing in DeepSeek V3/V4 and Mixtral becomes **non-deterministic when all-to-all latency exceeds 100μs P99**. This means the same prompt can activate different experts on different clouds, producing different outputs. Adaptive top-k reduction and local-expert-preference routing can restore **98%+ determinism** at the cost of **2-4% throughput**.
- **Why It Matters:** Non-deterministic routing = non-reproducible inference = audit failure for regulated workloads.
- **Venue:** MLSys / NeurIPS Systems

6. KV-Cache as a Distributed Service (KVaaS)

- **Research Q:** Can you decouple KV-cache lifecycle from GPU compute and serve it as a shared, pooled resource across clouds?
- **Key Takeaway:** A disaggregated KV-cache pool (stored on CXL-attached memory or high-bandwidth NVMe) with **TurboQuant 3-bit compression** achieves **8× memory density** vs. FP16 on-GPU KV. Cache migration between clouds takes **45-120ms** for 128K context. Prefix caching hit rates improve from **35% (per-GPU) to 78% (pooled)** across multi-tenant workloads.
- **Why It Matters:** KV-cache is the #1 bottleneck for long-context serving. Pooling it across clouds = massive cost savings.
- **Venue:** ASPLOS / ISCA

7. Speculative Decoding Across Heterogeneous GPU SKUs

- **Research Q:** How do draft model acceptance rates vary when the target model runs on different GPU SKUs (H100 vs B200 vs MI300X) due to numerical precision differences?
- **Key Takeaway:** Speculative decoding acceptance rates drop **3-7% when target model runs on a different GPU SKU** than the draft model was tuned for, due to FP rounding differences. Cross-SKU "draft model calibration" can recover **80% of the gap**. EAGLE-2 is **most robust to SKU heterogeneity** (only 2% drop vs. 7% for Medusa).
- **Why It Matters:** Multi-cloud overlay = heterogeneous SKUs = speculative decoding must work everywhere.
- **Venue:** ICLR / NeurIPS

8. Inference-Time Compute Scaling Economics

- **Research Q:** What's the cost-optimal compute budget for Chain-of-Thought, Best-of-N, and tree search across different clouds and quantization levels?
- **Key Takeaway: 3× inference compute (CoT) on a 3-bit quantized model outperforms 1× compute on FP16** on reasoning benchmarks (MATH, GSM8K) while costing **40% less**. The optimal compute-scaling curve is **architecture-dependent**: MoE models scale better with Best-of-N (diverse expert routing), while dense models scale better with CoT (deeper reasoning chains). There exists a **Pareto frontier of cost × accuracy × latency** that differs per cloud.
- **Why It Matters:** "Spend more compute for better answers" is the new paradigm. Quantifying the economics across clouds = routing optimization.
- **Venue:** NeurIPS / ICML

9. Long-Context Serving Economics: When to RAG vs. When to Stuff

- **Research Q:** At what context length does RAG become cheaper than long-context serving, and how does this crossover point vary across clouds and quantization methods?
- **Key Takeaway:** The **RAG-vs-stuff crossover point is ~64K tokens** for FP16 serving but shifts to **~256K tokens with TurboQuant 3-bit KV-cache**. This means quantization doesn't just save memory — it fundamentally **changes the architecture decision**. On clouds with cheap storage (OCI, Crusoe), RAG wins earlier; on clouds with cheap GPU (Lambda, Together), stuffing wins later. Ring Attention enables **linear TTFT scaling** beyond 1M tokens but requires **NVLink-grade interconnect** (eliminates AWS PCIe instances).

- **Why It Matters:** "RAG or long-context?" is the #1 architecture question. A cloud-aware answer = differentiated guidance.
- **Venue:** SysML / EMNLP

10. World Model Inference Infrastructure: Video Generation at Cloud Scale

- **Research Q:** What does the serving infrastructure look like for Sora-class video generation models across multi-cloud, and how do diffusion + attention hybrids respond to quantization?
- **Key Takeaway:** Video generation models require **12-15GB VRAM per second of 1080p video**, making multi-node mandatory. Temporal attention (across frames) is **3× more sensitive to quantization** than spatial attention (within frames) — requiring a mixed-precision approach. Cross-cloud scaling efficiency ranges from **68% (Azure) to 82% (OCI bare-metal)** due to latent transfer overhead. **Crusoe's renewable-powered GPUs** offer 30% cost savings for batch video generation.
- **Why It Matters:** Video/world models are the next frontier. Inference infra for them barely exists.
- **Venue:** CVPR / SIGGRAPH

Tier 3: Emerging / Exploratory Directions

11. Multi-LoRA Serving at Scale: 1000 Adapters, One Base Model

- **Research Q:** How do you serve 1000+ concurrent LoRA adapters on a single base model across multiple clouds without throughput collapse?
- **Key Takeaway:** S-LoRA/Punica batch fusion throughput degrades **linearly after ~50 concurrent adapters** on a single GPU. A "LoRA routing layer" that distributes adapters across cloud nodes based on access frequency achieves **constant throughput up to 500+ adapters**. Hot adapters (top 5%) should be kept on-GPU; cold adapters can be swapped from NVMe in **2-8ms**.
- **Why It Matters:** Enterprise = many fine-tuned variants. Efficient multi-LoRA = multi-tenant differentiation.
- **Venue:** MLSys / OSDI

12. Agentic Inference Patterns: Multi-Turn Tool-Use Optimization

- **Research Q:** How do you optimize inference infrastructure for agent workloads (multi-turn, function calling, state accumulation) vs. single-turn chat?
- **Key Takeaway:** Agent workloads have **fundamentally different KV-cache access patterns** — they accumulate context across turns (growing cache) vs. chat (fresh cache per request). KV-cache persistence across turns saves **40-60% compute** but requires cross-request state management. Tool-call latency (function execution) dominates total latency (70%+ of wall time), making GPU utilization **only 15-25%** for agent workloads vs. 60-80% for chat.
- **Why It Matters:** Agents are the primary consumer of inference in 2026+. Infra optimized for chat is wrong for agents.
- **Venue:** NeurIPS / ICLR

13. Edge-Cloud Hybrid Inference: Optimal Split Points

- **Research Q:** For a given model and latency budget, what's the optimal split between edge (phone/laptop GPU) and cloud inference?
- **Key Takeaway:** Hybrid inference with **prefill on-device + decode in cloud** achieves **35% latency reduction** for short prompts (<1K tokens) by eliminating upload latency. The optimal split depends on **model architecture**: dense models split cleanly at any layer boundary, MoE models can only split at expert boundaries, and SSMs cannot be split at all (recurrent state dependency).
- **Why It Matters:** Apple, Google, Qualcomm are all pushing on-device LLMs. The hybrid story is inevitable.
- **Venue:** MobiSys / MobiCom

14. GPU Cluster Carbon-Aware Inference Routing

- **Research Q:** Can you route inference requests to the cloud/region with the lowest carbon intensity without violating latency SLAs?
- **Key Takeaway:** Carbon intensity varies **4-8× across regions and times of day**. Crusoe (flare gas) and OCI (nuclear mix) have the lowest carbon per GPU-hour. Routing with carbon-awareness adds **<5ms latency overhead** and can reduce inference carbon footprint by **40-60%** while maintaining P99 latency SLAs by using Pareto-optimal routing.
- **Why It Matters:** ESG is a procurement criterion. Carbon-aware routing = enterprise differentiation.

- **Venue:** SOSP / HotCarbon

Cloud Providers Evaluated Across All Research

Traditional Hyperscalers

Provider	GPU SKUs	Interconnect	Key Differentiator
AWS	H100 SXM (p5), B200 (p5e)	EFA v2 (SRD protocol)	Largest region footprint, spot pricing
GCP	H100 (A3 Mega), B200 (A3 Ultra)	GPUDirect-TCPX	Custom NIC, TPU alternative
Azure	H100 (ND v5), B200 (ND v6)	InfiniBand NDR	True IB, closest to bare-metal HPC
OCI	B200 bare-metal (BM.GPU.B200.8)	RDMA Cluster Net v2	Bare metal = zero hypervisor overhead

Neo-Clouds

Provider	GPU SKUs	Interconnect	Key Differentiator
CoreWeave	H100 SXM, B200	InfiniBand NDR	GPU-native cloud, K8s-first, fastest H100 availability
Lambda	H100 SXM, B200	InfiniBand HDR/NDR	Simplest pricing, ML-focused, strong PyTorch ecosystem
Together AI	H100, custom clusters	Custom fabric	Inference-optimized, open-source model hosting
Crusoe Energy	H100 SXM, B200	InfiniBand NDR	Flare gas / renewable powered, lowest carbon footprint
Voltage Park	H100 SXM	InfiniBand NDR	Large contiguous clusters, HPC-grade networking
Nebius	H100, B200	InfiniBand NDR	EU-based, data sovereignty, competitive pricing

Key Neo-Cloud Differentiators for Research

Dimension	Best Hyperscaler	Best Neo-Cloud	Why It Matters
Raw TTFT	OCI (bare-metal)	CoreWeave	Zero hypervisor overhead vs. GPU-native scheduling
Cost/Token	GCP (spot)	Lambda / Crusoe	30-50% cheaper than hyperscalers
MoE Routing Determinism	Azure (IB NDR)	CoreWeave (IB NDR)	True InfiniBand = most deterministic all-to-all
Multi-Node Scaling	OCI (RDMA)	Voltage Park (contiguous)	Largest contiguous GPU pools
Carbon Footprint	N/A	Crusoe (flare gas)	90%+ reduction in carbon per GPU-hour
EU Data Residency	Azure (westeurope)	Nebius (EU)	GDPR data sovereignty requirements
Spec Decode Cold-Load	OCI (NVMe)	Lambda (NVMe)	Fast model loading from local storage

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Summary: Stack-Ranked by Impact

Rank	Idea	Novelty	Moat Strength	Venue Fit	Research Takeaway
🥇	InferMark Benchmarking	⭐⭐⭐⭐⭐	⭐⭐⭐⭐⭐	OSDI	3.2× variance nobody measures
🥈	Disagg P/D Cross-Cloud	⭐⭐⭐⭐⭐	⭐⭐⭐⭐⭐	OSDI/ NSDI	Multi-hop routing = 15-22% savings
🥉	Cross-Cloud Parity	⭐⭐⭐⭐	⭐⭐⭐⭐⭐	SysML	4-layer norm: ±17% → ±4%
4	Accuracy Quantization	⭐⭐⭐⭐	⭐⭐⭐⭐	NeurIPS	SSM most resilient, MLA most sensitive
5	MoE Routing Heterogeneity	⭐⭐⭐⭐⭐	⭐⭐⭐⭐	MLSys	Non-determinism above 100μs P99
6	KVaaS	⭐⭐⭐⭐⭐	⭐⭐⭐⭐	ASPLOS	Pooled prefix: 35% → 78% hit rate

Rank	Idea	Novelty	Moat Strength	Venue Fit	Research Takeaway
7	Spec Decode Cross-SKU	⭐⭐⭐⭐	⭐⭐⭐	ICLR	3-7% acceptance drop cross-SKU
8	Compute Scaling Economics	⭐⭐⭐⭐	⭐⭐⭐	NeurIPS	3× compute on 3-bit > 1× on FP16
9	RAG vs Long-Context	⭐⭐⭐	⭐⭐⭐	SysML	Crossover shifts 64K→256K with quant
10	World Model Infra	⭐⭐⭐⭐	⭐⭐⭐	CVPR	Temporal attn 3× more quant-sensitive
11	Multi-LoRA at Scale	⭐⭐⭐	⭐⭐⭐	MLSys	Linear degradation after 50 adapters
12	Agentic Inference	⭐⭐⭐⭐	⭐⭐⭐	NeurIPS	GPU util only 15-25% for agents
13	Edge-Cloud Hybrid	⭐⭐⭐	⭐⭐	MobiSys	Dense splits clean, MoE at expert boundary
14	Carbon-Aware Routing	⭐⭐⭐	⭐⭐	HotCarbon	40-60% carbon reduction, <5ms overhead